Explicit symmetric Padé kernels and exact odd-zeta cancellation

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Abstract

We identify the differential equations behind the moment data and construct explicit symmetric rational kernels with Barnes-integral representations. Finite rational linear conditions cancel every unwanted zeta coefficient. An exact term ratio and tail bound yield a saddle-point limit for normalized positive sums. Two kernel families and a finite search over subsidiary conditions are computed with complete gcd reduction. The construction produces pure-target forms, including several small finite forms for $\zeta(5)$, but does not establish a sequence of nonzero primitive forms tending to zero. The all-odd irrationality theorem is not proved.

1 Which differential equation is actually available?

The existing coefficients are defined by

$$\Delta_{N,a}(X) = \det[X - H^{(s)}_{a+i+j}]_{0\le i,j<N} = A_{N,a}X + B_{N,a}, \quad Q = \operatorname{lcm}(1,\dots,a+2N-2)^s,$$

$$U = Q^N A_{N,a}, \qquad V = Q^N B_{N,a}, \qquad g = \gcd(|U|,|V|), \qquad u = U/g.$$

An empty lcm is one. These formulas define u arithmetically; no linear differential equation or single Barnes integral for the gcd quotient has been established.

There is an exact equation for the moment-generating data. For $|z|<1$, absolute convergence gives

$$\sum_{n\ge0} H^{(s)}_n z^n = \frac{\operatorname{Li}_s(z)}{1-z}, \qquad \operatorname{Li}_s(z) = \sum_{n\ge1}\frac{z^n}{n^s}.$$

With $\theta = z\,d/dz$, we have $\theta^s \operatorname{Li}_s = z/(1-z)$, and hence

$$((1-z)\theta - 1)\theta^s \operatorname{Li}_s = 0. \tag{1}$$

For $s \ge 2$, its indicial exponents are

$$\begin{array}{c|c} 0 & 0 \text{ (s times)},\ 1 \\ 1 & 0,1,\dots,s-2,\ s-1,\ s-1 \\ \infty & 0 \text{ ($s+1$ times)} \end{array}$$

Indeed, the indicial polynomials are respectively $\rho^s(\rho-1)$, a nonzero multiple of $\rho(\rho-1)\cdots(\rho-s+1)(\rho-s+1)$, and a nonzero multiple of ρ^{s+1} after setting $z = 1/w$. These exponents do not determine an integral period lattice.

2 The exact arithmetic price of subtracting a counterterm

Let $\operatorname{den}(q) > 0$ denote a reduced rational denominator. For rational x, c, elementary denominator arithmetic gives

$$\operatorname{den}(x-c) \mid \operatorname{den}(x)\operatorname{den}(c), \qquad \operatorname{den}(x) \mid \operatorname{den}(x-c)\operatorname{den}(c). \tag{2}$$

The first follows by subtraction; the second by $x = (x-c)+c$. Consequently, subtracting a rational counterterm with denominator m can reduce the denominator of x by at most a factor m. An integer counterterm leaves it unchanged. Lean verifies these statements.

Euler–Maclaurin expansions can sharply estimate analytic remainders. To use a subtraction in an integer linear form, its rational denominator must still enter (2). A nonrational logarithmic counterterm adds another period unless its coefficient is independently canceled. Merely introducing and subtracting an intermediate scale does not alter the original reduced coefficient.

3 A concrete Barnes family with subsidiary conditions

Fix $r, n \geq 1$, put $s = 2r+1$, $A = 2r$, and choose $0 \leq m \leq rn$. These kernels have a different arithmetic budget from the single-shift determinants; the earlier lower bound is not asserted for them. Define

$$R_{r,n,m}(t) = \frac{\prod_{v=1}^{m}(t-v)(t+n+v)}{\prod_{j=0}^{n}(t+j)^{A}}. \tag{3}$$

This is a proper rational function of decay at least t^{-2r}. Away from the apparent gamma poles its exact gamma-ratio representation is

$$R_{r,n,m}(t) = \frac{\Gamma(t)^{A+1}\Gamma(t+n+m+1)}{\Gamma(t-m)\Gamma(t+n+1)^{A+1}}.$$

Both sides continue as the same rational function. Direct cancellation gives

$$R_{r,n,m+1}(t) = [t(t+n) - (m+1)(n+m+1)]R_{r,n,m}(t), \tag{4}$$

$$\frac{R_{r,n,m}(t+1)}{R_{r,n,m}(t)} = \frac{t^{A+1}(t+n+m+1)}{(t-m)(t+n+1)^{A+1}}, \tag{5}$$

where the second identity is used away from zeros and poles. These are actual recurrences for the rational kernel.

Proposition 1. *For $0 < c < 1$,*

$$\Lambda_{r,n,m} := -\sum_{k=1}^{\infty} R'_{r,n,m}(k) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\left(\frac{\pi}{\sin \pi t}\right)^2 R_{r,n,m}(t)\,\mathrm{d}t. \tag{6}$$

Moreover $\Lambda_{r,n,m} \in \mathbb{Q} + \mathbb{Q}\zeta(3) + \cdots + \mathbb{Q}\zeta(2r+1)$, with explicitly computable rational coefficients.

Proof. The series converges absolutely by the decay of R'. The integrand in (6) has residue $R'(k)$ at each positive integer. Close the upward vertical line to the right along half-integer rectangles. The orientation is clockwise; the remaining edges vanish by rational decay and the exponential decay of $\sin(\pi t)^{-2}$. This proves the sign and normalization of (6).

Write $R = \sum_{j=0}^{n}\sum_{k=1}^{A} c_{j,k}(t+j)^{-k}$. Its reflection symmetry $R(-n-t) = R(t)$ gives $c_{n-j,k} = (-1)^k c_{j,k}$. Differentiating the finite expansion and using $\sum_{\ell \geq 1}(\ell+j)^{-q} = \zeta(q) - H_j^{(q)}$ gives

$$\Lambda = \sum_{q=2}^{A+1}(q-1)\left(\sum_{j} c_{j,q-1}\right)\zeta(q) - \sum_{j=0}^{n}\sum_{k=1}^{A} k c_{j,k} H_j^{(k+1)}.$$

Even q vanish by the reflection identity. The coefficients follow from the finite Taylor formula

$$c_{j,k} = \frac{1}{(A-k)!}\,\frac{d^{A-k}}{dt^{A-k}}[(t+j)^A R(t)]\Big|_{t=-j}.$$

□

Take r block lengths $m_1, \dots, m_r$, and denote their coefficients of $\zeta(2q+1)$ by $a_{q,j}$. Choose rational, then integer, weights satisfying

$$\sum_{j=1}^{r} w_j a_{q,j} = 0 \qquad (1 \le q < r).$$

There is a nonzero rational solution because there are $r-1$ equations. The target coefficient must separately be nonzero; the finite audits check both rank and this nonvanishing. The resulting form is

$$L = \sum_j w_j \Lambda_{r,n,m_j} = a\zeta(s) + b.$$

For consecutive block lengths, the combined kernel is $R_{r,n,m_{\min}}(t)P(t(t+n))$ with $\deg P \le r-1$. Thus the subsidiary conditions define a concrete polynomial deformation.

A second audited family replaces (3) by

$$S_{r,n,m}(t) = \frac{(2t+n)^\epsilon \prod_{v=1}^{m}(t-v)(t+n+v)}{\prod_{j=0}^{n}(t+j)^s}, \qquad \epsilon \equiv n \pmod 2,$$

with $2m+\epsilon \le s(n+1)-2$. Then $S(-n-t) = -S(t)$, so $\sum_{k\ge1} S(k)$ again has only odd zeta terms and a rational constant. Its total simple-pole residue is zero by proper decay; this accounts for the convergent first-order pole terms.

There is also an analytic nonvanishing mechanism. At positive integers, $S(k) = 0$ for $k \le m$ and $S(k) > 0$ for $k > m$: beyond the numerator zeros every displayed factor is positive. Lean verifies these inequalities for the actual rational kernel and the positive-value conclusion for any of its convergent sums. If the subsidiary weights have one sign and are not all zero, the combined integer summands have that sign and are strictly nonzero beyond all block lengths. Their sum is therefore nonzero. Constructing such weights with adequate size bounds remains an arithmetic requirement; positivity is not assumed for arbitrary nullvectors.

3.1 The functional Padé bridge

Let $c_{j,q}$ be the partial-fraction coefficients of S. Define

$$P_q(z) = \sum_{j=0}^{n} c_{j,q} z^{n-j}, \qquad P_0(z) = \sum_{j=0}^{n}\sum_{q=1}^{s} c_{j,q} \sum_{\ell=1}^{j} \frac{z^{n-j+\ell}}{\ell^q}.$$

All these polynomials have degree at most n. Reindexing absolutely convergent series for $|z| < 1$ gives the exact functional identity

$$\sum_{q=1}^{s} P_q(z)\,\mathrm{Li}_q(z) - P_0(z) = z^n \sum_{k\ge1} S(k) z^k = O(z^{n+m+1}). \tag{7}$$

The zero order follows from the first m numerator zeros. Reflection gives $P_q(1) = 0$ for even q, and proper decay gives $P_1(1) = 0$. Hence the logarithmic term $P_1(z)\,\mathrm{Li}_1(z)$ tends to zero at $z = 1$: $P_1(z) = O(1-z)$ and $\mathrm{Li}_1(z) = -\log(1-z)$. The additional odd-coefficient conditions cancel the remaining unwanted values. This supplies an actual functional Padé problem at 0 and subsidiary conditions at 1, rather than assuming an identification with the polylogarithm differential system.

4 What the cancellations achieve, and their remaining cost

At $r = 2, n = 1$, choosing the two weights $-23, -145$ for block lengths $2, 1$ gives, with $T = t(t+1)$,

$$K(t) = \frac{-23T^2 + 39T + 14}{T^4} = -\frac{17}{t^3} + \frac{14}{t^4} + \frac{17}{(t+1)^3} + \frac{14}{(t+1)^4}.$$

Consequently

$$-K'(t) = -\frac{51}{t^4} + \frac{56}{t^5} + \frac{51}{(t+1)^4} + \frac{56}{(t+1)^5}, \qquad -\sum_{k\geq 1} K'(k) = 112\zeta(5) - 107.$$

Lean verifies the rational identities and the corresponding `HasSum` identity for the actual real zeta value. It also verifies $112\zeta(5) - 107 \geq 5$. Thus exact lower-period cancellation alone does not give smallness.

The two families were checked at 92 weight/index/profile choices. A separate search enumerated 2140 distinct nonzero-target two-kernel forms for $\zeta(5)$, with complete gcd reduction. The indices are $2, 4, 8, 12, 20$, with respectively $21, 66, 231, 496, 1326$ distinct candidates. For the first profile the full combined kernel is explicitly

$$F(t) = 601088S_{2,2,2}(t) + 37632S_{2,2,3}(t) = \frac{256(147t^2 + 294t + 143)(t-1)(t-2)(t+3)(t+4)}{t^5(t+1)^5(t+2)^5}.$$

Its partial fractions give $\sum_{k\geq 1} F(k) = 91776\zeta(5) - 95165 > 0$, with the entire $\zeta(3)$ coefficient canceled. The next best primitive form is $1216512\zeta(5) - 1261435$. Both absolute values are certified below one. All signs and finite comparisons use exact rational zeta enclosures. Finite smallness is not a limit theorem.

If D clears the denominators of a, b and $g = \gcd(|Da|, |Db|)$, the necessary combined estimate is

$$0 < |L_{\mathrm{prim}}| = \frac{D}{g}|L| \leq \frac{D}{g}\sum_j |w_j|\,|\Lambda_j| \longrightarrow 0.$$

A Barnes saddle estimate must therefore be paired with bounds on the subsidiary-condition weights and the exact arithmetic normalization. These bounds have not been proved to win for every odd target.

The general contour, gamma-ratio, indicial, functional Padé, and saddle-limit arguments are paper proofs here. Lean verifies the centering budget, finite period-coordinate elimination, the explicit $\zeta(5)$ series identities, and integer-kernel positivity. No all-index target nonvanishing or winning arithmetic estimate is asserted.

There are specialized alternatives to a general Siegel–Shidlovskii statement for G-functions. Fischler's multipoint multiplicity argument [2] proves the infinitely-many-odd-values result, whereas Rivoal [1] shows that the direct general G-function analogue is false. Recent Hurwitz-zeta Padé asymptotics [3] assume determinantal nondegeneracy and a regular equilibrium regime; they do not assert the integer coefficient estimate above. The explicit families here provide objects on which to test such methods, without assuming their missing arithmetic conclusion.

5 Analytic control of the positive summands

This calculation concerns the sums before the subsidiary-condition combination and final primitive normalization. Restrict to even n, fix $0 < \beta < s/2$, and let the integers $m = m_n$ satisfy $m_n/n \to \beta$. Put

$$F_{n,m}(k) = \frac{(n!)^s}{(m!)^2} S_{r,n,m}(k), \qquad \mathcal{A}_{n,m} = \sum_{k>m} F_{n,m}(k).$$

The summands are positive and have the exact ratio

$$\frac{F_{n,m}(k+1)}{F_{n,m}(k)} = \frac{k^{s+1}(k+n+m+1)}{(k-m)(k+n+1)^{s+1}}.$$

Define, for $x > \beta$,

$$\begin{aligned}\Phi_{s,\beta}(x) = (s+1)[x\log x - (x+1)\log(x+1)] \\ + (x+1+\beta)\log(x+1+\beta) \\ - (x-\beta)\log(x-\beta) - 2\beta\log\beta.\end{aligned}$$

There is a unique maximizing point x_β. Indeed, with $v = x/(x+1)$, the equation $\Phi' = 0$ becomes

$$G(v) := \beta(1+v^{s+1}) - \sum_{j=1}^{s} v^j = 0.$$

For $0 < v < 1$, $G'(v) = \beta(s+1)v^s - \sum_{j=1}^{s} jv^{j-1} < 0$, because $\sum jv^{j-1} \geq s(s+1)v^s/2$ and $2\beta < s$. Also $G(\beta/(1+\beta)) > 0$ and $G(1) < 0$. The same argument, with β replaced by $m/(n+1)$, proves unimodality of the exact discrete summands.

Here is a uniform tail bound, so the infinite sum introduces no missing exponential factor. Set

$$E = s(n+1) - 2m - 2, \quad B = 2m^2 + sn(n+1), \quad K = \max\{2m, m+1, \lceil B/E \rceil\}.$$

For large n, $E > 0$ and $K = O(n)$. For real $t \geq K$,

$$\frac{d}{dt}\log[t^2 S(t)] \leq -\frac{E}{t} + \frac{B}{t^2} \leq 0.$$

To verify this, use $\sum_{v=1}^{m}(t-v)^{-1} \leq m/t + 2m^2/t^2$, $\sum_{v=1}^{m}(t+n+v)^{-1} \leq m/t$, and $\sum_{j=0}^{n}(t+j)^{-1} \geq (n+1)/t - n(n+1)/t^2$. Thus $F(k) \leq F(K)(K/k)^2$ for $k \geq K$, and

$$\max_{k>m} F(k) \leq \mathcal{A}_{n,m} \leq (2K+1)\max_{k>m} F(k).$$

Finally the elementary factorial estimate $\log(q!) = q\log q - q + O(\log(q+2))$, obtained by bounding the sum of $\log j$ between its adjacent integrals, gives $n^{-1}\log F_{n,m}(\lfloor nx \rfloor) \to \Phi_{s,\beta}(x)$ locally uniformly for $x > \beta$. The exact ratio locates the peak at $k/n \to x_\beta$. Consequently

$$\lim_{\substack{n\to\infty \\ n \text{ even}}} \frac{1}{n}\log \mathcal{A}_{n,m_n} = \Phi_{s,\beta}(x_\beta). \tag{8}$$

In particular the certified rate for $s = 5$, $\beta = 1$ lies in $(-4.970794, -4.970793)$. This proves exponential decay of those normalized positive sums. The scripts certify 44 negative saddle-rate intervals and 22 finite positive-sum enclosures against independent period enclosures.

For $\beta = 1$, decay follows at every odd weight, not just the tested weights. For $x > 1$, $B(x) := x\log x - (x+1)\log(x+1) < 0$, and hence

$$\Phi_{s,1}(x) = \Phi_{3,1}(x) + (s-3)B(x) \leq \Phi_{3,1}(x).$$

The certified base maximum is less than -1.99. Thus, for each fixed odd $s \geq 3$, $(n!)^{s-2}\sum_{k\geq 1} S_{r,n,n}(k) \leq e^{-1.99n}$ for all sufficiently large even n.

5.1 Completion of the lower-period analytic step

For a fixed $s = 2r+1$, use the r normalized sums with block lengths $m_j = n - j$, $0 \leq j < r$. Each has rate limit $\Phi_{s,1}(x_1)$, so all are bounded by $e^{-1.99n}$ for sufficiently large even n. Write their lower-period coefficient matrix as $C_n \in M_{r-1,r}(\mathbb{Q})$. Dimension gives a nonzero $w_n \in \ker C_n$. These weights are computable: when C_n has full row rank, use the signed maximal minors $w_{n,j} = (-1)^j \det \widehat{C_n^j}$. Laplace expansion with a repeated row proves $C_n w_n = 0$. Otherwise rational row reduction and a free coordinate produce a kernel vector. Put $M_n = \sum_j |w_{n,j}| > 0$ and $\widehat{w}_{n,j} = w_{n,j}/M_n$. Then, exactly,

$$C_n\widehat{w}_n = 0, \qquad \sum_j |\widehat{w}_{n,j}| = 1, \qquad L_n := \sum_j \widehat{w}_{n,j}\mathcal{A}_{n,n-j} = a_n\zeta(s) + b_n,$$

$$|L_n| \le \sum_j |\widehat{w}_{n,j}|\,|\mathcal{A}_{n,n-j}| \le e^{-1.99n}.$$

This completes lower-period cancellation and preserves analytic decay. Lean verifies the rational kernel-vector existence, normalization, cancellation, norm bound, and resulting convergence implication.

This normalization does not itself give small integer forms. If $a_n \neq 0$, multiplying both rational coefficients by the same nonzero rational does not change the reduced denominator of b_n/a_n; Lean verifies this invariance. One must still prove nonvanishing and $(D_n/g_n)|L_n| \to 0$ after the full integer normalization.

References

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[2] S. Fischler, *Shidlovsky's multiplicity estimate and Irrationality of zeta values*, 2016. https://arxiv.org/abs/1609.09770.

[3] A. Kandaian, *Multipoint Padé Approximation of the Hurwitz Zeta Function and a Riemann–Hilbert Steepest Descent Analysis*, preprint, 2026. https://arxiv.org/abs/2602.08399.